

Financial Transactions Modeler Case Study

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Rules

- **Time limit and submission:** The maximum time allowed is **3 hours**. The time starts when you receive the email containing the case study and the accompanying data tables. At the end of 3 hours, please export your code and any output, including comments, in whichever format you prefer and email the file back to the same Swiss Re email address you received the case study from. Submissions received more than 3 hours after the case study was sent will not be considered.
- **Use of AI:** The use of artificial intelligence tools, including but not limited to generative AI coding assistants, chatbots, automated reasoning tools, or AI-based code generation tools, is not permitted. The purpose of this case study is to assess your own ability to structure, code, and explain a quantitative model.
- **Permitted tools:** You may use the programming language of your choice. Solutions implemented in Excel or any spreadsheet application will not be considered. You may, however, use a spreadsheet application to inspect the data or to create CSV/XLSX files for data ingestion.
- **Completeness:** Candidates are not expected to fully solve every question within the allotted time. Incomplete submissions will also be reviewed, particularly where the code structure, modelling approach, comments, and assumptions are clear.

Assessment Focus

Candidates are not expected to have prior specialized knowledge of insurance products. The objective is to assess the ability to understand the provided specification and translate it into a quantitative model.

The thought process is more important than producing fully production-ready code. Please include comments in your code explaining your analytical approach, modelling choices, assumptions, and any simplifications. Comments will be considered as part of the assessment. If a question is unclear, or if you are unable to fully solve part of the problem, please add comments explaining your doubts, assumptions, and the clarifying questions you would ask in a live coding or modelling discussion. This will contribute positively to the assessment.

The assessment will focus on:

- Correct implementation of product mechanics based on the specification
- Coding clarity and robustness
- Quality of comments and explanation
- Ability to identify assumptions, limitations, and clarification points

You should be prepared to discuss your code, modelling choices, assumptions, and thought process in subsequent interview rounds.

Case Study

The objective of this case study is to model the cashflows and Present Values (PV) of the Variable Annuity portfolio offered by Company XYZ. You should build a monthly projection model that estimates fees, withdrawals, guarantee claims, and net profitability under the base case and selected shock scenarios.

Assume the model projects monthly over 40 years, with a pricing date of January 1, 2026. For this case study, model policies deterministically using the supplied fund paths, discount factors, mortality rates, and lapse rates. Do not run stochastic simulations.

Company XYZ offers a Variable Annuity. Each policyholder's Account Value (AV) is initially invested 50% in the Equity Fund and 50% in the Bond Fund, and is rebalanced back to this 50% / 50% allocation on each policy anniversary.

The following data are provided in the accompanying tables:

- Table 1: In-force policy details
- Table 2: Projected fund values
- Table 3: Discount curve
- Table 4: Mortality assumptions
- Table 5: Lapse assumptions

Part A - Base Product Projection

Build a monthly projection model for the base product over the next 40 years.

The base product includes a Return of Premium Guaranteed Minimum Death Benefit, or GMDB. The fee for the base product is a Mortality and Expense Allowance (MEA) fee, of 1.5% per annum of Account Value. This fee includes the cost of the base Return of Premium death benefit, plus all expenses.

Use the following projection conventions:

- Account Value: Each policy starts with the Account Value shown in Table 1. Account Value evolves according to the returns of the Equity Fund and Bond Fund in Table 2, reflecting the initial 50% / 50% allocation and annual rebalancing on each policy anniversary.
- Fees: The MEA fee is charged quarterly in arrears at the end of March, June, September, and December, based on the average Account Value for the quarter. The MEA fee is revenue to the insurer and reduces Account Value dollar-for-dollar.
- Fee floor: Fees cannot reduce Account Value below zero. If Account Value is insufficient to cover the full fee, collect only the available Account Value and set Account Value to zero. No fee shortfall is recovered.
- GMDB Benefit Base: For the base product, the GMDB Benefit Base is initially equal to Initial Premium in Table 1. Unless otherwise stated in later parts, it remains unchanged after issue.
- GMDB claim: On death, the death benefit payable is the greater of Account Value and GMDB Benefit Base. The insurer claim cost is the excess of the guaranteed death benefit over Account Value.

$$\text{GMDB Claim} = \max(\text{GMDB Benefit Base} - \text{Account Value}, 0)$$

- Mortality and lapse: Mortality and lapse assumptions are provided as annual rates. Convert them to monthly rates. For mortality rates, use attained age at the start of each projection month, rounded down to the last completed integer age. Use MaleMort for policies with Sex = M and FemaleMort for policies with Sex = F. Apply lapse rates by completed policy duration; duration 0 applies during the first policy year and the 10+ lapse rate applies from duration 10 onward.
- Deterministic policyholder projection: For each policy in Table 1, assume an initial policyholder count of 1. Do not run stochastic simulations of individual deaths or lapses. Mortality and lapse rates should be applied deterministically to this initial policyholder count over the projection period. This means that the projected number of in-force policyholders may become fractional over time. Fractional policyholder counts are acceptable and should be interpreted as the best estimate of surviving and non-lapsed policyholders.
- Projection timing: Use a monthly projection. Start with policies in force at the beginning of the month; apply monthly investment return; deduct fees if the month is a fee month; apply death and lapse decrements at month-end; calculate claims arising from deaths.
- Discounting: To calculate Present Values (PV), use the monthly discount factors in Table 3. All cashflows occurring in a projection month should be discounted using the discount factor corresponding to that month.
- Simplifications: Ignore taxes, commissions, capital costs, statutory reserves, hedging costs, additional expenses, dynamic policyholder behaviour, stochastic scenarios, and regulatory constraints. Model all policies independently and aggregate at the end. Partial-month effects may be ignored.

Your code should output projected monthly values for at least:

- Policyholder count
- Deaths
- Lapses
- Aggregate AV
- Aggregate GMDB Benefit Base
- Aggregate MEA fees
- Aggregate GMDB claims

The code should also calculate:

- PV of projected MEA fees
- PV of projected GMDB claims
- Net PV, defined as PV MEA Fees - PV GMDB Claims

Based on your results, is the base product profitable?

Part B - Roll-up GMDB Rider

Company XYZ decides to include a 6% Annual Roll-up GMDB rider for policies with HasRollUpGMDBRider = Y.

- At each policy anniversary, the existing GMDB Benefit Base increases by 6% while the policy remains in force.
- The additional Roll-up rider fee is 0.30% per annum of the GMDB Benefit Base, charged quarterly in arrears.
- The Roll-up rider fee is revenue to the insurer and reduces Account Value dollar-for-dollar. Fees cannot reduce Account Value below zero.
- The GMDB claim formula remains the same as in Part A, but using the updated GMDB Benefit Base.

Amend your code from Part A to include this policy feature.

In addition to part A, your code should output projected monthly values for at least:

- Aggregate Roll-up GMDB rider fees

The code should also calculate:

- PV of Roll-up GMDB rider fees
- Updated net PV
- Net PV of the rider, defined as net PV with rider - net PV without rider (from part A)

Is the Roll-up GMDB rider profitable?

Part C - GLWB Rider

Company XYZ also offers a 5% Guaranteed Lifetime Withdrawal Benefit (GLWB) rider for policies with HasGLWB Rider = Y.

- The annual guaranteed withdrawal amount is 5% of the GLWB Benefit Base, paid monthly in arrears.
- Withdrawals start on the policy's 10th anniversary.
- Assume all eligible policyholders fully utilize the GLWB rider. Excess withdrawals above the guaranteed amount are outside the scope of this case study.
- The GLWB Benefit Base is equal to Initial Premium at issue. Scheduled GLWB withdrawals do not reduce the GLWB Benefit Base; for this case study, the GLWB Benefit Base remains equal to Initial Premium throughout the projection.
- GLWB withdrawals reduce Account Value dollar-for-dollar, subject to Account Value not going below zero.
- If the scheduled withdrawal exceeds the available Account Value at the withdrawal date, the insurer pays the shortfall as a GLWB claim.

`GLWB Claim = max(Scheduled Withdrawal - Account Value before withdrawal, 0)`

- After the withdrawal and any GLWB claim payment, Account Value is floored at zero. Once Account Value has reached zero, the insurer continues to pay the guaranteed withdrawal amount while the policyholder remains alive and in force.
- No further MEA fees or rider fees are collected from a policy once Account Value has reached zero.
- Scheduled GLWB withdrawals also reduce the GMDB Benefit Base, where applicable. The reduction is applied proportionally to the reduction in Account Value. If Account Value before withdrawal is zero, no further proportional reduction is applied.
- The GLWB rider fee is 1.25% per annum of the GLWB Benefit Base, charged quarterly in arrears. The fee is revenue to the insurer and reduces Account Value dollar-for-dollar, subject to the fee floor described above.
- If a policy has both the Roll-up GMDB rider and the GLWB rider, GMDB roll-ups terminate on the policy's 10th anniversary, when withdrawals are set to start.

Update your code to include the GLWB feature.

In addition to part B, your code should output projected monthly values for at least:

- Scheduled withdrawals
- Aggregate GLWB Benefit Base
- Aggregate GLWB rider fees
- Aggregate GLWB claims

The code should also calculate:

- PV of GLWB rider fees
- PV of GLWB claims
- Net PV of the GLWB rider, defined as net PV with rider - net PV without rider (from part B)

Is the GLWB rider profitable?

Part D - Equity Shock Scenario

Assess the effect of a -20% Equity Fund shock at inception.

- Reduce the Equity Fund value at inception by 20%.
- Project subsequent Equity Fund values using the same relative returns as in the base scenario.
- Assume that the Bond Fund value is unchanged.

Re-run the model and compare against the base scenario. Comment on the impact on:

- PV fees
- PV claims
- Net PV
- Product profitability
- Riders profitability, where applicable
- Is the assumption of an unchanged Bond Fund value realistic?

Part E - Interest Rate Shock Scenario

Assess the effect of a -100bps interest rate shock at inception.

- Assume the Bond Fund has a modified duration of 5 years.
- Apply the shock to the Bond Fund at inception using the modified duration approximation.
- Project subsequent Bond Fund values using the same relative returns as in the base scenario.
- Assume that the Equity Fund value is unchanged.

Re-run the model and compare against the base scenario. Comment on the impact on:

- PV fees
- PV claims
- Net PV
- Product profitability
- Rider profitability, where applicable
- Is the assumption of an unchanged Equity Fund value realistic?

Part F - Model Review and Client Questions

Please list any additional items you think Swiss Re or Company XYZ should consider before using this model for pricing, risk management, or product design.

Examples may include:

- Modelling assumptions that should be validated
- Product features that require clarification
- Data limitations
- Sensitivities or stresses that should be tested
- Hedging or capital considerations
- Practical implementation risks

Please also list any clarifying questions.